The Bare Binary Biconditional: Its Quantum-Calculus Correspondence and Implications for the Equal-Weight Envariance Gap

A Short Cross-Disciplinary Synthesis

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Abstract

This paper synthesizes established results connecting bare binary symmetry, half-probability, asymptotic half-frequency, Lebesgue measure, integration, and symmetric binary quantum measurement. Zurek's envariance program seeks to explain the equal-weight assignment through physical symmetry, while published critiques isolate the transition from state symmetry to probability symmetry as the disputed step. A probability law ν_p on $\{0,1\}$ is called *bare* here when exchanging the labels 0 and 1 leaves the law invariant. If $\tau(0)=1$ and $\tau(1)=0$, then

$$\tau_*\nu_p=\nu_p \quad\Longleftrightarrow\quad p=\frac{1}{2}.$$

Independent identical repetition gives the Bernoulli product measure $\mu_p=\nu_p^{\otimes\mathbb{N}}$ on $\Omega=\{0,1\}^{\mathbb{N}}$. Under the dyadic map

$$\phi(\omega)=\sum_{k=1}^{\infty}\omega_k 2^{-k},$$

its pushforward is Lebesgue measure λ on $[0,1]$ exactly when $p=1/2$. Writing

$$S_n(\omega)=\sum_{k=1}^{n}\omega_k,$$

the complete correspondence within the Bernoulli family is

$$\tau_*\nu_p=\nu_p \quad\Longleftrightarrow\quad p=\frac{1}{2} \quad\Longleftrightarrow\quad \phi_*\mu_p=\lambda \quad\Longleftrightarrow\quad \mu_p\left(\left\{\omega:\lim_{n\to\infty}\frac{S_n(\omega)}{n}=\frac{1}{2}\right\}\right)=1.$$

The final condition entails that both 0 and 1 occur infinitely often. Informally, and with the bare and independent-repetition qualifications made explicit: 0 and 1 states imply the $1/2$ law, while convergence to $1/2$ implies that both states occur without bound.

The pushforward identity links the binary system directly to calculus because Lebesgue integration over $[0,1]$ is integration over fair infinite binary records in a second representation.

Existing binary quantum measurements already supply the physical two-outcome behavior and recorded sequences used in this synthesis; no additional measurement is proposed or required. Everett relates repeated measurement records to product-measure and frequency reasoning. For the paper's bounded physical-bareness case, Zurek supplies exchange symmetry, Bell excludes local predetermined accounts of the relevant correlations, Pironio and collaborators certify genuine randomness under stated assumptions, and Colbeck and Renner limit improved prediction under theirs. These results contribute different parts of the physical case; none alone selects $1/2$. The explicit bareness premise

supplies that selection: within the bounded physical description, a structureless binary alternative is represented by an exchange-invariant probability law.

For a specified symmetric state and binary measurement, the Gleason-Busch trace rule also yields equal probabilities. The present synthesis gives an independent route: under the explicit bareness premise, the bounded physical case is represented by a swap-invariant binary law, which uniquely fixes $1/2$, and that same value uniquely produces Lebesgue measure and the continuum used by integration. This is a symmetric binary account, not a general derivation of unequal quantum weights. The author's conjecture about observation remains a separate proposed research direction.

1 Introduction and Exact Scope

This inquiry began while reasoning about binary behavior toward infinity and about real-world scenarios whose observed proportions approach $1/2$. In the corresponding fair independent model, the normalized frequencies converge to $1/2$ almost surely. The value $1/2$ is not a third outcome: each recorded result remains 0 or 1, and each symbol occurs infinitely often in a typical infinite record. This suggested a focused question: can binary quantum behavior and continuum calculus be related through the mathematics of half-probability at infinity?

The central mathematical components already exist. Binary product measure, dyadic expansion, normal numbers, Lebesgue integration, Hilbert-space probability measures, quantum measurement, envariance, and algorithmic randomness were developed in different fields and for different purposes [1-18]. The contribution attempted here is a synthesis of those components into one explicit structure. It does not claim a new theorem about product measure, a new quantum measurement, or a new general probability rule.

The paper uses the author's original biconditional as one statement rather than as two unrelated observations:

> *Within a bare* $[0, 1]$ *binary system, the availability of the 0 and 1 states fixes the probability law at* $1/2$*; in an infinite independent record governed by that law, convergence to* $1/2$ *entails that 0 and 1 both occur infinitely often.*

The biconditional was formulated before any of the literature reviewed here was located. The sections that follow identify which established results the statement already relied on and where each had been developed. That ordering bears on Section 7: the statement was reachable without prior access to any of the component literatures, which is consistent with the finding that those literatures do not cite one another at the points joined here. No priority over any component theorem is claimed.

The word *bare* is load-bearing. It does not mean merely that an observer is ignorant of a bias. In the formal statement below, it means that the one-trial probability law contains no preferred outcome label: exchanging 0 and 1 leaves the law unchanged. A biased binary law is therefore not a counterexample; it violates the stated symmetry condition. Independence is a separate assumption introduced only when passing from one trial to an infinite record. Bare symmetry by itself does not imply independent repetition.

The paper keeps four levels distinct:

1. **Established mathematics:** swap-invariant binary probability, Bernoulli product measure, almost-sure frequency, dyadic pushforward, and the integral identity.
2. **Physical and foundational literature:** binary quantum measurements, recorded outcomes, Hilbert-space measures, envariance, and certified randomness.
3. **Interpretive synthesis:** the same value $1/2$ selected by a bare binary law is the unique Bernoulli parameter that produces Lebesgue measure under dyadic coding.

4. **Author's conjecture:** observation may make a quantum object behave as if it is at an asymptotic boundary, while unobserved behavior may remain continuum-like.

Only the first two levels establish the cited results. The third states the cross-disciplinary connection argued in this paper. The fourth neither supports nor follows from the synthesis; it is included because it motivated the search and identifies the intended next research question.

2 The Bare Binary Statement

2.1 Direction 1: bare symmetry fixes the one-trial law

Let

$$\nu_p(\{1\}) = p, \qquad \nu_p(\{0\}) = 1 - p, \qquad 0 \le p \le 1,$$

be a probability law on the two-element space $\{0, 1\}$. Define the exchange map

$$\tau(0) = 1, \qquad \tau(1) = 0.$$

The pushforward law satisfies

$$(\tau_* \nu_p)(\{1\}) = \nu_p(\tau^{-1}\{1\}) = \nu_p(\{0\}) = 1 - p.$$

Consequently,

$$\tau_* \nu_p = \nu_p \quad \Longleftrightarrow \quad p = 1 - p \quad \Longleftrightarrow \quad p = \frac{1}{2}.$$

This is Direction 1 of the biconditional in exact probability language. A bare binary law is not defined only by the names 0 and 1. It is the two-outcome law for which the outcomes are exchangeable at the level of the law. There is then no free probability parameter: normalization and exchange symmetry force the two singleton probabilities to be $1/2$.

This formulation also identifies the boundary of the claim. If a physical preparation, selection rule, or other condition makes one outcome probabilistically preferred, its law is not invariant under τ, and it is not bare in this stated sense. The paper does not infer that every system with two possible recorded values is fair.

2.2 Direction 2: independent records, frequency, and recurrence

To discuss a growing record, add independent identical repetition. Let

$$\Omega = \{0, 1\}^{\mathbb{N}}, \qquad \mu_p = \nu_p^{\otimes \mathbb{N}},$$

and write

$$S_n(\omega) = \sum_{k=1}^{n} \omega_k.$$

For a finite word $a_1 \ldots a_n$, the corresponding cylinder set has measure

$$\mu_p([a_1 \dots a_n]) = p^{a_1+\cdots+a_n}(1-p)^{n-(a_1+\cdots+a_n)}.$$

The Bernoulli law of large numbers gives

$$\frac{S_n}{n} \longrightarrow p \quad \text{for } \mu_p\text{-almost every } \omega.$$

For the bare law selected in Section 2.1,

$$\frac{S_n}{n} \longrightarrow \frac{1}{2} \quad \text{for } \mu_{1/2}\text{-almost every } \omega.$$

Independence also gives a precise meaning to the author's phrase *either outcome can occur at any time*. For every finite history $h = [a_1 \dots a_n]$ and either $a \in \{0, 1\}$,

$$\mu_{1/2}(\omega_{n+1} = a \mid h) = \frac{1}{2}.$$

Thus neither outcome is blocked by any finite record. This does not guarantee which result occurs on a particular trial; it states that every finite history leaves both next outcomes available with equal conditional probability.

Convergence to $1/2$ entails unbounded occurrence of both symbols. If 1 occurred only finitely many times, then $S_n/n \to 0$. If 0 occurred only finitely many times, then $S_n/n \to 1$. Therefore, on every record for which $S_n/n \to 1/2$, both 0 and 1 occur infinitely often.

In the mathematical sections, *at infinity* means along an unbounded sequence of trials. It does not mean that both outcomes occur simultaneously, that there is a final trial located at infinity, or that any one trial is guaranteed to produce a specified result.

The converse must not be overstated: the mere fact that both symbols occur infinitely often does not force their limiting frequency to be $1/2$. Direction 2 concerns records governed by the fair independent product law, or records already known to converge to half-frequency. It is not a claim about every infinite binary sequence.

The qualification *almost surely* is equally important. A probability-one claim permits exceptional sequences of measure zero. The synthesis concerns the measure and its typical records, not a claim that every formally describable sequence is balanced.

2.3 Exact completion and asymptotic frequency are different facts

At depth n, the fair binary product space has 2^n cylinder sets. Each has measure 2^{-n}, so their total is exactly

$$2^n 2^{-n} = 1.$$

No limiting argument is needed to make the finite-depth partition complete. Every depth already partitions the whole probability space.

The frequency statement is different. The ratio S_n/n generally fluctuates at every finite n and approaches $1/2$ only asymptotically for typical records. Thus:

- total measure is exactly 1 at every finite dyadic depth;
- individual outcomes are always exactly 0 or 1; and
- relative frequency converges to $1/2$ along a typical infinite record.

Keeping these facts separate prevents the completed measure space from being mistaken for a temporal process that must first perform infinitely many steps.

3 From the Binary Product to Lebesgue Calculus

3.1 The dyadic pushforward

Map an infinite binary record to the unit interval by

$$\phi(\omega) = \sum_{k=1}^{\infty} \omega_k 2^{-k}.$$

The two binary expansions of each dyadic rational create a countable ambiguity. That ambiguity is a null set for the measures at issue and does not alter the conclusions.

Under the fair law, every depth-n cylinder has probability 2^{-n}. Its dyadic image is an interval of length 2^{-n}. Since the cylinders and dyadic intervals generate their respective measurable spaces, agreement at every finite depth gives

$$\phi_* \mu_{1/2} = \lambda,$$

where λ is Lebesgue measure on $[0, 1]$. Łomnicki and Ulam give the equal-weight product construction and its connection with the binary representation of the interval [2].

The converse distinguishes $1/2$ from every other Bernoulli parameter. Under μ_p, digit frequency is p almost surely. Borel's normal-number result states that Lebesgue-almost every point of $[0, 1]$ has binary digit frequency $1/2$ [1].[1] If $p \neq 1/2$, the set on which frequency converges to p has full μ_p-measure after dyadic coding but zero Lebesgue measure. The induced measure is therefore singular with respect to λ.

The exact measure-theoretic statement is

$$\boxed{\tau_* \nu_p = \nu_p \quad \Longleftrightarrow \quad p = \frac{1}{2} \quad \Longleftrightarrow \quad \phi_* \mu_p = \lambda \quad \Longleftrightarrow \quad \mu_p\left(\left\{\omega : \lim_{n\to\infty} \frac{S_n(\omega)}{n} = \frac{1}{2}\right\}\right) = 1.}$$

This resolves the direction of the construction without assuming uniformity twice. Exchange invariance first selects the fair law on the primitive two-element space. Independent repetition then forms its product measure. The dyadic aggregate of that product measure is Lebesgue measure. Conversely, requiring the dyadic aggregate of a Bernoulli product law to be Lebesgue selects the same fair parameter. The final equivalence follows from the Bernoulli law of large numbers: within this family, almost-sure half-frequency holds exactly when $p = 1/2$.

3.2 Integration over binary records

The pushforward identity has an immediate integral form. For every Lebesgue-integrable function $f : [0, 1] \to \mathbb{R}$,

[1] The full text of Borel's 1909 paper could not be obtained for purposes of this paper; the DOI resolves to a Springer Nature record from which the text was not available. Borel is cited for the well-known normal-number result where that attribution is required, using the bibliographic record, the surrounding historical literature, and the explicit discussion in Łomnicki and Ulam (1934). No page-specific quotation or further claim about the substance of Borel's original paper is made here.

$$\boxed{\int_0^1 f(x)\,d\lambda(x) = \int_\Omega f\left(\sum_{k=1}^{\infty} \omega_k 2^{-k}\right) d\mu_{1/2}(\omega).}$$

The continuum integral and the integral over fair binary records are two representations of the same measure calculation.

Binary-valued functions are also the elementary units in the standard construction of the Lebesgue integral. For a measurable set A,

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A \end{cases} \qquad \text{and} \qquad \int \mathbf{1}_A\,d\lambda = \lambda(A).$$

Finite weighted sums of indicator functions are simple functions. Nonnegative measurable functions are integrated through increasing simple approximations, and integrable signed functions are obtained from their positive and negative parts. For ordinary continuous functions on a compact interval, this agrees with the usual definite integral.

The link to calculus is therefore exact rather than pictorial. Fair binary product measure does not merely resemble repeated subdivision. Through ϕ, it is the standard interval measure over which integration is performed.

3.3 The single synthesized object

The two directions and the calculus link can now be read as one structure:

1. bare exchange symmetry fixes the binary law at $1/2$;
2. independent repetition produces typical records whose frequencies converge to $1/2$ and in which both outcomes recur without bound;
3. dyadic aggregation turns that product measure into Lebesgue measure; and
4. integration over the interval is consequently integration over the same fair binary product space in another representation.

The established theorems in these steps are not claimed as new. The paper's claim is that their combination gives a direct quantum binary-to-calculus identification and a precise interpretation of the author's biconditional.

4 Replacing the Zeno Picture of Calculus

Zeno's dichotomy is commonly invoked as a description of the limiting behavior later formalized by calculus. That framing is inaccurate.

Written as an argument, the dichotomy requires the following steps:

1. traversing an interval can be described by an unending list of successively smaller subintervals;
2. an unending list of subtasks cannot be completed; therefore
3. the interval cannot be traversed.

The second statement is an unsupported premise, not a consequence of the first. The purported paradox therefore fails as logic before any appeal to calculus. Convergent series and integration provide mathematical countermodels to the unsupported premise, but they are not historically or logically generated by Zeno's argument.

The binary-Lebesgue construction supplies the clearer replacement description. At each finite dyadic depth, all cylinders already have total measure 1. In the completed product space, every coordinate is a definite 0 or 1; under the fair law, the aggregate is exactly Lebesgue measure and supports the standard integral. A physically recorded binary result likewise does not occupy a partial outcome between 0 and 1. The calculus behavior is described by exact measure and its binary representation, not by an imagined traveler trapped in an unfinished sequence.

5 Binary Quantum Measurement and Probability

5.1 What Gleason and Busch already establish

Gleason characterizes countably additive probability measures on the closed subspaces of a separable Hilbert space of dimension at least three. Under his assumptions, such a measure has trace form

$$\Pr(P) = \operatorname{tr}(\rho P)$$

for a positive trace-one operator ρ [3]. Busch extends the structural conclusion to generalized effects and includes the two-dimensional case [11].

For a specified symmetric state and binary measurement, this formalism does yield $1/2$. Let $P_0 + P_1 = I$, and suppose a transformation U exchanges the two outcomes,

$$P_1 = UP_0U^\dagger,$$

while leaving the state invariant,

$$U\rho U^\dagger = \rho.$$

Then cyclicity of the trace gives

$$\operatorname{tr}(\rho P_1) = \operatorname{tr}(\rho U P_0 U^\dagger) = \operatorname{tr}(U^\dagger \rho U P_0) = \operatorname{tr}(\rho P_0).$$

Normalization makes both probabilities $1/2$.

This is already a complete calculation once the state ρ, the measurement, and the probability representation are supplied. It is not, however, the same question as explaining why probabilities attach to the measurement alternatives in that form or why physical symmetry warrants the corresponding probability symmetry. Gleason assumes a normalized additive measure on projections; Busch assumes the analogous measure on effects. They constrain its representation. The envariance debate concerns the physical and interpretive bridge to equal probability, not an inability to perform the trace calculation.

5.2 Existing records and the bounded physical-bareness case

No additional binary quantum measurement is required by this paper. The literature already treats measurements with two recorded alternatives and the statistics of repeated records. Under the independent-repetition

model used here, repeated identical preparations give independent records. Everett describes measurement as an interaction correlating a measured system with a corresponding observer record, and he relates the weights of successive records to a product measure and asymptotic relative frequency [18]. That use of records and product measure is directly relevant here without requiring this paper to adopt every interpretive claim associated with Everett's framework.

The paper uses *physical bareness* in a bounded, operational sense. The specified binary alternatives must be exchange-symmetric, and the available physical description must contain no outcome-distinguishing predictive structure of the kinds excluded by the cited results. Four parts of the literature contribute to that case:

1. Zurek supplies the exchange symmetry for the specified equal-weight alternatives through envariance [5,6].
2. Bell shows that locally predetermined variables cannot reproduce the correlations in his theorem [4].
3. Pironio and collaborators show that an observed Bell-inequality violation can certify genuine output randomness under explicit physical and independence assumptions [12].
4. Colbeck and Renner argue, under their quantum and free-choice assumptions, that an extension cannot improve predictive power [13].

These results do different work. Bell excludes a broad and important class of local predetermined accounts, not every conceivable underlying account. Pironio and collaborators certify min-entropy, not necessarily a uniform raw output. Colbeck and Renner's conclusion is conditional on their assumptions. None of the three independently selects $1/2$, and Zurek's transition from state symmetry to probability symmetry is the disputed step examined in Section 6.

Bareness premise. This paper assumes that when the binary alternatives are exchange-symmetric and the cited results exclude outcome-distinguishing structure in the bounded senses just stated, the one-trial law is exchange-invariant. This premise is not derived from Bell, Pironio, or Colbeck and Renner; those results support the premise but do not complete it. It performs the same formal work as the probability-relabeling premise in Zurek's argument. Its stated basis is excluded outcome-distinguishing structure together with exchange symmetry, rather than state symmetry alone.

Wheeler's proposal that registered binary answers may be foundational and that continuum descriptions may be derived is a close conceptual precedent [17]. It is used as historical framing, not as a premise in the measure-theoretic proof.

5.3 Why algorithmic randomness is relevant

The paper reserves *bare* for the one-trial probability law and for a physical system argued to instantiate that law-level symmetry. Algorithmic randomness has a separate role: it supplies a rigorous notion of effective typicality for an individual sequence through statistical null tests. It does not create the fair physical law or provide a second definition of bareness.

Nies and Scholz extend Martin-Löf randomness to infinite quantum-state systems and prove an associated law-of-large-numbers result [14]. Bhojraj develops related notions of quantum algorithmic randomness, including a Schnorr-random class satisfying a law of large numbers [15]. Bhojraj's later measurement framework connects measurement in computable bases with Martin-Löf-random binary records and the corresponding frequency law [16].

These sources have a specific role in the synthesis: they bridge the ensemble-level fair product measure to rigorous typicality claims about individual records. In the fair setting, an algorithmically random record has half-frequency and therefore contains both outcomes infinitely often. Algorithmic randomness does not prove that a physical preparation is fair, and a simple half-frequency condition is weaker than full normality or Martin-Löf randomness.

6 The Equal-Weight Envariance Gap

6.1 Zurek's symmetric case

For an entangled composite state with equal-magnitude coefficients, Zurek shows that swapping two states of the measured subsystem can be undone by a counterswap acting on the environment, leaving the joint state unchanged [5]. In the 2003 argument, the connection to probability includes an additional premise: when the alternatives are swapped, their probabilities are correspondingly relabeled. Equal swappability, normalization, and that premise produce equal probability.

Zurek's expanded treatment gives several versions of the argument and extends the equal case to unequal coefficients by fine-graining, additivity, normalization, and continuity [6]. In the equal case, envariance establishes a physical symmetry of the state. The disputed point is how that symmetry licenses symmetry of the probability assignment.

6.2 Published critiques and neighboring routes

Schlosshauer and Fine isolate assumptions connecting outcome states, local properties, correlated partners, and invariance of one subsystem's probabilities under transformations of the other [7]. Their central concern is that symmetry of the state does not by itself establish the required behavior of probabilities.

Mohrhoff similarly argues that envariance can establish symmetry once probabilities are associated with the alternatives, but does not by itself explain why probabilities, outcomes, or records enter the account [9].

Barnum gives a no-signalling version that avoids Zurek's original probability-relabeling assumption by applying no-signalling in both directions and using perfect correlation [8]. It offers a distinct bridge to equal weighting, while still employing physical probability conditions. Deutsch supplies a separate decision-theoretic route from a set of rationality and physical assumptions to the standard weights [10]. These are neighboring solutions to the same foundational territory, not premises of the present measure construction.

6.3 What the synthesis contributes to the gap

The synthesis offers an alternative bridge across the disputed equal-weight step. In place of Zurek's probability-relabeling premise it supplies the bareness premise of Section 5.2, and it embeds the resulting value in an independently established measure structure:

$$\begin{aligned}\text{bare binary law} &\iff \tau_*\nu_p = \nu_p \iff p = \frac{1}{2} \\ &\iff \phi_*\mu_p = \lambda \\ &\iff \mu_p\left(\left\{\omega : \lim_{n\to\infty} \frac{S_n(\omega)}{n} = \frac{1}{2}\right\}\right) = 1.\end{aligned}$$

The first equivalence is the paper's explicit definition of bare probability symmetry. The second is an elementary consequence of normalization on a two-element space. The third is the classical dyadic product-measure result [1,2], and the fourth is the Bernoulli frequency theorem. The resulting $1/2$ is therefore not introduced solely as a relabeling convention inside the envariance argument. It is the unique law of a label-symmetric binary probability space, the unique Bernoulli law whose dyadic aggregate is the standard continuum measure, and the unique parameter in that family whose typical records have half-frequency.

The physical bridge is the bareness premise stated explicitly in Section 5.2. Under that premise, exchange symmetry together with the bounded exclusions supported by the cited results warrants representing the one-trial law by $\tau_* \nu_p = \nu_p$. The cited results support but do not derive this premise. Applying the premise selects $1/2$; Bell alone does not.

The two premises can be compared directly. Zurek's premise moves from a symmetry of the joint state to a corresponding relabeling of probabilities, and the critiques in Section 6.2 locate the difficulty exactly there. The bareness premise does not appeal to state symmetry. It appeals to the absence of outcome-distinguishing structure in the bounded physical description, and that absence is supported by results constraining what such a description can contain [4,12,13]. The premise remains a premise, and the exclusions are bounded rather than exhaustive. The difference is where it is exposed: to challenges about the scope of those exclusions, rather than to the state-to-probability transition that the published critiques target.

This accounts for the omitted measure-theoretic part of the symmetric binary case: it states the independent premise, identifies its bounded physical support, proves which probability value it permits, and shows why that value has an independently distinguished relationship to calculus. State symmetry alone is not silently renamed probability symmetry. The Gleason-Busch framework supplies the probability representation, while the bareness premise supplies the independent route from the supported physical case to the exchange-invariant probability law.

Accordingly, the paper does not claim a new derivation of the complete unequal-weight rule, nor does it claim that the dyadic map is a physical operation performed by an apparatus. Its result is bounded to the symmetric binary case. Existing binary measurements supply the physical outcomes; no further measurement is needed.

7 Why the Connection Was Available but Unmade

Every mathematical component in Sections 2 and 3 is established, and most of the physical and foundational components in Sections 5 and 6 are decades old. That raises an immediate question: why was the combined binary, Lebesgue, calculus, and envariance structure still available to synthesize? The sources reviewed here show that its components were distributed across communities addressing different problems and, in the supplied core papers, the relevant communities did not cite one another across the crucial divide.

The components belong to:

- **Real analysis and calculus:** dyadic expansion, convergent series, and definite integration.
- **Measure theory:** product measures, pushforwards, singularity, indicators, simple functions, and Lebesgue measure.
- **Probability:** Bernoulli trials, almost-sure convergence, product distributions, and relative frequency.
- **Algorithmic randomness and computability:** effective null sets, Martin-Löf randomness, Schnorr randomness, and laws of large numbers.

- **Hilbert-space measure theory:** Gleason-type representations of normalized additive measures.
- **Quantum foundations and measurement theory:** recorded outcomes, entanglement, envariance, locality, and competing accounts of probability.
- **Information theory:** binary records and certified randomness.
- **Decision theory and philosophy of probability:** the interpretation and operational use of probability weights.

A full-text and reference-list check of the supplied papers found that the five envariance papers and responses [5-9] cite neither Borel's normal-number work [1] nor Łomnicki and Ulam's product-measure construction [2]. Conversely, the three supplied quantum algorithmic-randomness papers [14-16] do not engage Zurek, the cited envariance responses, Deutsch's derivation, or the equal-weight derivation debate.[2] These verified non-citations do not prove that no earlier paper made the synthesis. They do show that the core literatures used here developed without direct cross-reference at the points joined by this paper.

The fields are not claimed to be equivalent, and no priority is claimed over any component theorem. The proposed contribution is the connection: one shared structure links bare binary symmetry, the fair product law, typical half-frequency, Lebesgue measure, calculus, and the symmetric equal-weight question. Its cross-disciplinary distribution explains how each component could be familiar while the complete chain remained unstated in the materials reviewed.

8 Why the Synthesis Matters

The supportable mathematical result is the exact chain

$$\tau_*\nu_p = \nu_p \quad \Longleftrightarrow \quad p = \frac{1}{2} \quad \Longleftrightarrow \quad \phi_*\mu_p = \lambda \quad \Longleftrightarrow \quad \mu_p\left(\left\{\omega : \lim_{n\to\infty} \frac{S_n(\omega)}{n} = \frac{1}{2}\right\}\right) = 1,$$

together with

$$\int_0^1 f\, d\lambda = \int_\Omega (f \circ \phi)\, d\mu_{1/2}.$$

This identifies one object at four levels: a bare binary law, its fair independent product, typical half-frequency, and the continuum measure used by integration.

That structure matters on its own. It explains why $1/2$, rather than an arbitrary parameter, occupies the boundary between an outcome-symmetric binary system and the standard measure of the continuum. It also places the almost-sure half-frequency and infinite recurrence of both outcomes inside the same construction rather than presenting them as disconnected coincidences.

Physically, existing binary quantum records provide a realized setting in which discrete outcomes, probability, repeated records, and symmetry are all active questions. The synthesis does not require a new experiment to create that setting. For the equal-weight envariance question, it supplies the explicit probability-law structure that was left contested. The binary-to-Lebesgue correspondence does not provide a second warrant for the

[2]The full text and reference lists of the copies used for this synthesis were checked on 29 July 2026. In [5-9], no citation to Borel or Łomnicki-Ulam was found. In [14-16], no engagement with Zurek, Schlosshauer and Fine, Barnum, Mohrhoff, Deutsch, or the Born-rule derivation debate was found. This is evidence about the supplied core sample, not an exhaustive priority search.

physical case; it shows that the value fixed by that structure is the same value uniquely distinguished within the Bernoulli family by the continuum measure.

9 Motivating Quantum Conjecture and Proposed Next Step

The following is the author's motivating conjecture, not a result of the preceding synthesis:

> ***Quantum observation conjecture.*** *Measurement causes a quantum object to behave as if it is at infinity, producing a definite binary record, whereas an unobserved object may continue to exhibit continuum-like behavior.*

The phrase *as if it is at infinity* has not yet been assigned a formal physical operation. The reviewed literature establishes neighboring but strictly narrower points: measurement can produce correlated records [18]; repeated records admit product-measure, frequency, and algorithmic-typicality analysis [2,14-16,18]; and symmetric quantum states support equal-weight arguments whose probability bridge remains a foundational subject [3,5-11]. None of these results proves the conjecture.

Its role is to explain why the synthesis was sought and to identify the single proposed follow-up: determine whether the observation language can be formalized into a testable distinction between definite binary records and continuum-level behavior. Until then it remains an explicitly labeled interpretive conjecture.

10 Conclusion

For a probability law on $\{0, 1\}$, invariance under exchange of the two outcomes is equivalent to the unique assignment $p = 1/2$. With independent identical repetition, that law produces a Bernoulli product measure whose typical records converge to half-frequency and contain each outcome infinitely often. Under binary expansion, the same product measure pushes forward exactly to Lebesgue measure on $[0, 1]$, and Lebesgue integration is therefore integration over that binary record space in another representation.

This unifies the author's two-direction biconditional without treating its directions as unrelated claims. Bare symmetry fixes the law; independent repetition gives the frequency behavior; and the dyadic map gives the calculus continuum. The finite-depth partitions are exactly complete at every stage even though record frequency is asymptotic.

In quantum foundations, Gleason and Busch show how a specified symmetric state and binary measurement yield equal trace probabilities. Everett connects measurement records with product-measure reasoning. Zurek derives a physical swap symmetry through envariance, while the published responses identify the move from state symmetry to probability symmetry as the contested equal-weight step. Zurek, Bell, Pironio and collaborators, and Colbeck and Renner contribute distinct, bounded parts of the physical bareness case. The explicit bareness premise then supplies the independent route to an exchange-invariant probability law and $1/2$. The same parameter is the unique Bernoulli parameter that generates the standard continuum measure and almost-sure half-frequency. This is a symmetric binary result, not a new general unequal-weight derivation.

The observer conjecture remains separate from every established claim in the paper. It is included only as the motivating and proposed next research direction.

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Appended Note: Preparation and Research Provenance

The author, Christopher M Struck, formulated the bare binary biconditional before this project began and without the assistance described below, identified its correspondence with the equal-weight step examined in Section 6, and determined the logic, scope, and claims of this paper. The observer-effect conjecture in Section 9 is likewise the author's.

Large language model assistance was used for literature search, reference retrieval, drafting, and the citation check reported in Section 7. Assistance was provided by Opus 5, Sonnet 5, Fable 5, Grok 4.5, and GPT 5.6. Those searches located the component results in pre-existing recorded work and did not locate the full cross-disciplinary biconditional as a single formal statement in the materials reviewed. The author verified the resulting attributions and retains responsibility for the framing and claims.